

✓ Capstone Project - Document Processing Agent

User: Rajkumar

Requirements:

- OCR for scanned PDFs
- Classification + Extraction using LLM
- Structured JSON output
- Human-in-the-loop
- LangGraph workflow

```
!apt-get install -y tesseract-ocr poppler-utils
!pip install langchain==0.2.16 langgraph pypdf pytesesseract pdf2image pillow openai
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tesseract-ocr is already the newest version (4.1.1-2.1build1).
poppler-utils is already the newest version (22.02.0-2ubuntu0.12).
0 upgraded, 0 newly installed, 0 to remove and 6 not upgraded.
Requirement already satisfied: langchain==0.2.16 in /usr/local/lib/python3.12/dist-packages (0.2.16)
Requirement already satisfied: langgraph in /usr/local/lib/python3.12/dist-packages (0.4.5)
Requirement already satisfied: pypdf in /usr/local/lib/python3.12/dist-packages (6.9.2)
Requirement already satisfied: pytesesseract in /usr/local/lib/python3.12/dist-packages (0.3.13)
Requirement already satisfied: pdf2image in /usr/local/lib/python3.12/dist-packages (1.17.0)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (11.3.0)
Requirement already satisfied: openai in /usr/local/lib/python3.12/dist-packages (2.28.0)
Requirement already satisfied: PyYAML>=5.3 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (6.0.3)
Requirement already satisfied: SQLAlchemy<3,>=1.4 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.0)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (4.0.0)
Requirement already satisfied: langchain-core<0.3.0,>=0.2.38 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.2.38)
Requirement already satisfied: langchain-text-splitters<0.3.0,>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.2.0)
Requirement already satisfied: langsmith<0.2.0,>=0.1.17 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.1.17)
Requirement already satisfied: numpy<2.0.0,>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.26.0)
Requirement already satisfied: pydantic<3,>=1 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.12.3)
Requirement already satisfied: requests<3,>=2 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.32.4)
Requirement already satisfied: tenacity!=8.4.0,<9.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (8.4.0)
Requirement already satisfied: langgraph-checkpoint<3.0.0,>=2.0.26 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.0.26)
Requirement already satisfied: langgraph-prebuilt<=0.1.8 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.1.8)
Requirement already satisfied: langgraph-sdk<=0.1.42 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.1.42)
Requirement already satisfied: xxhash<4.0.0,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (3.5.0)
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (24.2)
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (4.12.1)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.28.1)
Requirement already satisfied: jiter<1,>=0.10.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.13.0)
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.3.1)
Requirement already satisfied: tqdm>4 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (4.67.3)
Requirement already satisfied: typing-extensions<5,>=4.11 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (4.15.0)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.5.0)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (25.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.4.1)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (6.0.0)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.2.0)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.17.0)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (3.11)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2026.0.1)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.0.7)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.16.0)
Requirement already satisfied: jsonpatch<2.0,>=1.33 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.33)
Requirement already satisfied: ormsgpack>=1.10.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.10.0)
Requirement already satisfied: orjson>=3.11.5 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (3.11.5)
Requirement already satisfied: requests-toolbelt<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (1.0.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (3.4.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.3.0)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (3.1.1)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from langchain==0.2.16) (2.5.0)
```

```
import os
from openai import OpenAI
from langchain_core.messages import HumanMessage
from langgraph.graph import StateGraph
from typing import TypedDict
from google.colab import files
from pdf2image import convert_from_path
```



```
import pytesseract
from pypdf import PdfReader
```

```
client = OpenAI(
    base_url='https://openrouter.ai/api/v1',
    api_key='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'
)
```

```
uploaded = files.upload()
```

Choose Files 19 files

```
bw_doc_1.pdf(application/pdf) - 312773 bytes, last modified: 3/17/2026 - 100% done
bw_doc_2.pdf(application/pdf) - 274613 bytes, last modified: 3/17/2026 - 100% done
bw_doc_3.pdf(application/pdf) - 285612 bytes, last modified: 3/17/2026 - 100% done
bw_doc_4.pdf(application/pdf) - 284236 bytes, last modified: 3/17/2026 - 100% done
bw_doc_5.pdf(application/pdf) - 288554 bytes, last modified: 3/17/2026 - 100% done
LoA1.pdf(application/pdf) - 311776 bytes, last modified: 3/17/2026 - 100% done
LOA2.pdf(application/pdf) - 311914 bytes, last modified: 3/17/2026 - 100% done
LOA3.pdf(application/pdf) - 367412 bytes, last modified: 3/17/2026 - 100% done
LOA4.pdf(application/pdf) - 319210 bytes, last modified: 3/17/2026 - 100% done
LOA5.pdf(application/pdf) - 269051 bytes, last modified: 3/17/2026 - 100% done
LOA6.pdf(application/pdf) - 791380 bytes, last modified: 3/17/2026 - 100% done
LOA7.pdf(application/pdf) - 813193 bytes, last modified: 3/17/2026 - 100% done
LOA8.pdf(application/pdf) - 1139882 bytes, last modified: 3/17/2026 - 100% done
LOA9.pdf(application/pdf) - 1364891 bytes, last modified: 3/17/2026 - 100% done
notice_1.pdf(application/pdf) - 562221 bytes, last modified: 3/17/2026 - 100% done
notice_2.pdf(application/pdf) - 543574 bytes, last modified: 3/17/2026 - 100% done
notice_3.pdf(application/pdf) - 549749 bytes, last modified: 3/17/2026 - 100% done
notice_4.pdf(application/pdf) - 534156 bytes, last modified: 3/17/2026 - 100% done
notice_5.pdf(application/pdf) - 530647 bytes, last modified: 3/17/2026 - 100% done
Saving bw_doc_1.pdf to bw_doc_1 (1).pdf
Saving bw_doc_2.pdf to bw_doc_2 (1).pdf
Saving bw_doc_3.pdf to bw_doc_3 (1).pdf
Saving bw_doc_4.pdf to bw_doc_4 (1).pdf
Saving bw_doc_5.pdf to bw_doc_5 (1).pdf
Saving LoA1.pdf to LoA1 (1).pdf
Saving LOA2.pdf to LOA2 (1).pdf
Saving LOA3.pdf to LOA3 (1).pdf
Saving LOA4.pdf to LOA4 (1).pdf
Saving LOA5.pdf to LOA5 (1).pdf
Saving LOA6.pdf to LOA6 (1).pdf
Saving LOA7.pdf to LOA7 (1).pdf
Saving LOA8.pdf to LOA8 (1).pdf
Saving LOA9.pdf to LOA9 (1).pdf
Saving notice_1.pdf to notice_1 (1).pdf
Saving notice_2.pdf to notice_2 (1).pdf
Saving notice_3.pdf to notice_3 (1).pdf
Saving notice_4.pdf to notice_4 (1).pdf
Saving notice_5.pdf to notice_5 (1).pdf
```

```
def extract_text(file_path):
    try:
        reader = PdfReader(file_path)
        text = ''
        for page in reader.pages:
            text += page.extract_text() or ''
        if len(text.strip()) > 50:
            return text
    except:
        pass
    images = convert_from_path(file_path, dpi=100, first_page=1, last_page=2)
    text = ''
    for img in images:
        text += pytesseract.image_to_string(img)
    return text

documents = []
for file in uploaded.keys():
    print(f'Processing: {file}')
    text = extract_text(file)
    documents.append({'file': file, 'text': text})

print(f'Processed {len(documents)} documents')
```

```
Processing: bw_doc_1 (1).pdf
Processing: bw_doc_2 (1).pdf
Processing: bw_doc_3 (1).pdf
Processing: bw_doc_4 (1).pdf
Processing: bw_doc_5 (1).pdf
```

```
Processing: LoA1 (1).pdf
Processing: LOA2 (1).pdf
Processing: LOA3 (1).pdf
Processing: LOA4 (1).pdf
Processing: LOA5 (1).pdf
Processing: LOA6 (1).pdf
Processing: LOA7 (1).pdf
Processing: LOA8 (1).pdf
Processing: LOA9 (1).pdf
Processing: notice_1 (1).pdf
Processing: notice_2 (1).pdf
Processing: notice_3 (1).pdf
Processing: notice_4 (1).pdf
Processing: notice_5 (1).pdf
Processed 19 documents
```

```
class State(TypedDict):
    file: str
    text: str
    result: str
    approved: bool
```

```
def process_doc(state):
    prompt = f"""
Return ONLY valid JSON:

{{
  "doc_type": "",
  "parties": [],
  "date": "",
  "summary": "",
  "confidence": 0
}}

Text:
{state['text'][:3000]}
"""

    response = client.chat.completions.create(
        model="openai/gpt-3.5-turbo",
        messages=[
            {"role": "user", "content": prompt}
        ],
        temperature=0
    )

    state["result"] = response.choices[0].message.content
    return state
```

```
def review(state):
    print(f"\nFile: {state['file']}")
    print(state['result'])
    decision = input('Approve? (yes/no): ')
    state['approved'] = decision.lower() == 'yes'
    return state
```

```
builder = StateGraph(State)
builder.add_node('process', process_doc)
builder.add_node('review', review)
builder.set_entry_point('process')
builder.add_edge('process', 'review')
graph = builder.compile()
```

```
results = []
for doc in documents:
    state = {'file': doc['file'], 'text': doc['text'], 'result': '', 'approved': False}
    results.append(graph.invoke(state))
results
```

